

Order of Operations and Percentages

Baldivis
Secondary College

Name: _____

Total Marks: /16

ANSWERS + MARK BREAKDOWN

Resources allowed: Calculator

Total Time: 16 mins

Weighting: 1%

* Overall loss of a total of $\frac{1}{2}$ mark for not putting units in final answer.

1. [5 marks]

Scoring in basketball depends on where a player shoots from. A successful shot from inside the three-point line scores two points; from outside the three-point line, it scores three points. Each successful free shot from the foul line scores one point.

The number of successful shots for two teams during a match is shown in the table

Number of successful shots	Titans	Giants
inside the three-point line	32	35
outside the three-point line	7	5
from a free throw	4	2

- a) Complete these statements to show the numbers and operations needed to calculate the total points scored by each team.

$$\text{Points scored by Titans} = 32 \times 2 + 7 \times 3 + 4 \times 1$$

$$\text{Points scored by Giants} = 35 \times 2 + 5 \times 3 + 2 \times 1$$

- b) Calculate the number of points each team scored

$$\begin{aligned} \text{Titans} &: 32 \times 2 + 7 \times 3 + 4 \times 1 \\ &= 64 + 21 + 4 \\ &= 89 \text{ points} \end{aligned}$$

$$\begin{aligned} \text{Giants} &: 35 \times 2 + 5 \times 3 + 2 \times 1 \\ &= 70 + 15 + 2 \\ &= 87 \text{ points} \end{aligned}$$

- c) Which team won and by how much?

Titans won by 2 points.

2. [2 marks]

Write each percentage as a decimal

a) 23%

0.23

b) 16.52%

0.1652

3. [2 marks]

Write each decimal as a percentage

a) 0.47

47% (1)

b) 0.0003

0.03% (1)

4. [1 mark]

Write $\frac{17}{20}$ as a percentage

$$\frac{17}{20} \times 100 = 85\% \quad (1)$$

5. [1 mark]

Write 24 out of 150 as a percentage

$$\frac{24}{150} \times 100 = 16\% \quad (1)$$

6. [1 mark]

Calculate 37% \$2400

$$0.37 \times 2400 = \$888 \quad (1)$$

or

$$\frac{37}{100} \times 2400$$

7. [4 marks]

Brock is a farmer who intends on increasing the amount of land he owns. He currently has 250 hectares. He would like 20% more land.

a) Calculate how much land Brock would have if he increased his land by 20%

$$0.2 \times 250 = 50 \text{ hectares} \quad (1)$$
$$\text{New total: } 250 + 50 = 300 \text{ hectares} \quad (1)$$

b) Brock was able to buy 40 hectares of land. Was this more or less than the amount that he wanted to get?

Less (1)

c) Does Brock need to purchase any more hectares to get the total amount of land he wanted? If so how much.

Yes, he needs to purchase 10 more hectares. (1)

~End of Test~